

Unit 2- Things and Connections



Objective:

After completing this course, learner will be able to :

- Use different tools for editing image
- Understand different type of image editing
- Working with multiple layers.
- Creating banner, logos, cards and Posters



Introduction to Control System

- Control system is a way of arranging and combining components in such a way that the desired output is obtained .
- Working of Controlled System
 - In the context of control system, a plant is a system to be controlled.
- The system can be controlled **manually** and **automatic**.



Features of a Control System

- Accuracy
- Sensitivity
- Energy savings
- Improved safety
- Noise
- Stability
- Bandwidth
- Speed
- Oscillation



Type of Control System

- There are three basic types control system that are:
 - Feed Forward Control System
 - Open Loop Control System
 - Closed Loop or Feedback Controls System



Feed Forward Control System

- Feed forward is a type of system which counters the changes in its surroundings.
- It is used to maintain the wanted status of the system.
- The biggest advantage of this system is that it does not permit the huge disturbance in the output.
- The objective of feed forward control, also known as predictive control, is to measure or predict and potential open-loop disturbances and compensate for them manually before the controlled variable deviates too far from the original set point.

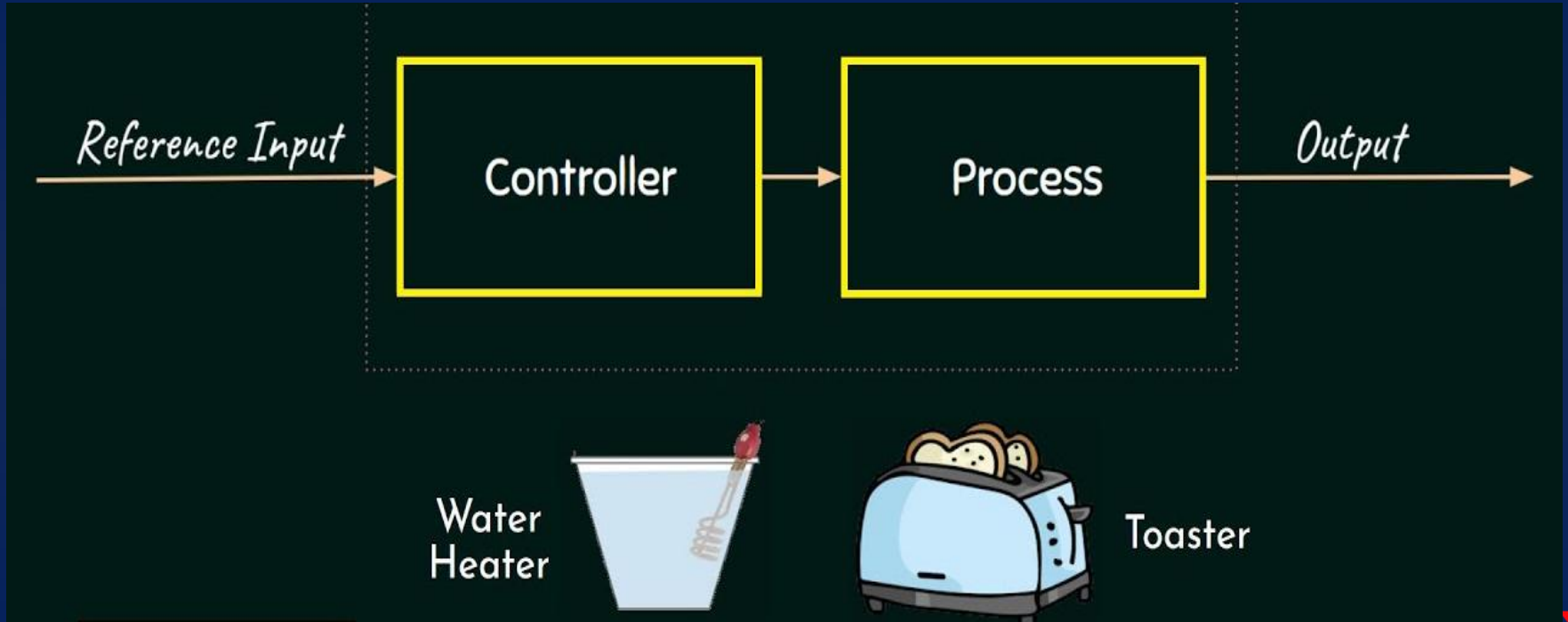


Open Loop Control System

- A control system in which the control action is totally independent of output of the system then it is called open loop control system.
- A **manual control system** is also known as open-loop control system.
- An open loop control system is also referred as **non-feedback control system**.
- Some other examples of open loop control system are:
 - Electric Hand Drier
 - Automatic Tea/Coffee Maker
 - Bread Toaster
 - Traffic Light Control System
 - Electric clothes dryer



Block diagram of Open loop system

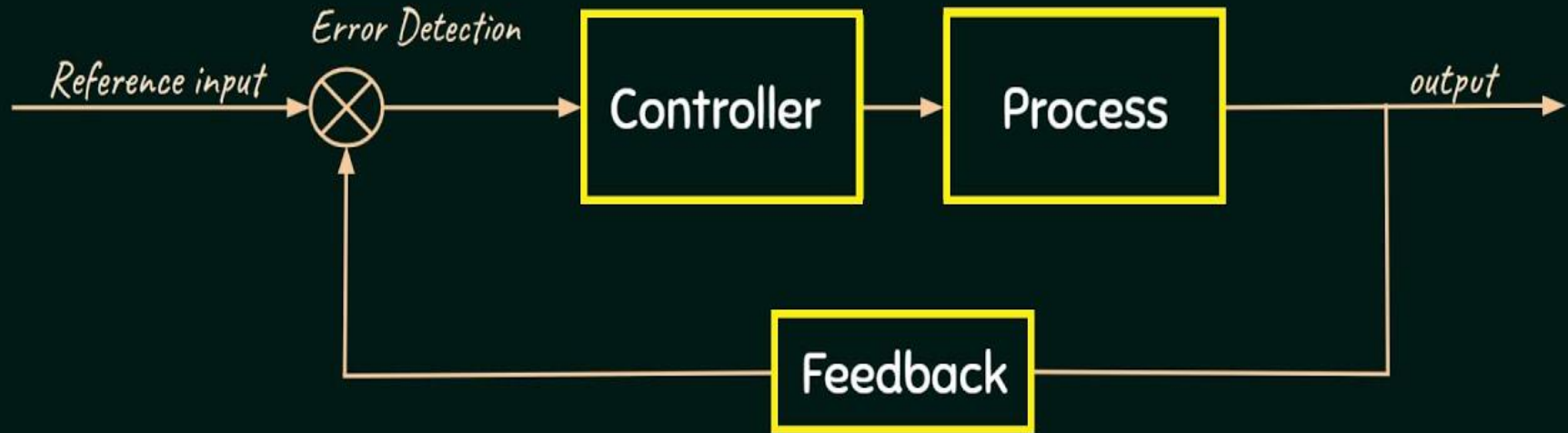


Closed Loop or Feedback Control System

- Closed loop control system is a control system where its control action depends on both of its input signal and output response.
- Closed loop control also known as feedback control system.
- Open loop control system can be converted into closed loop control system by providing a feedback. This feedback automatically makes the suitable changes in the output due to external disturbance.
- In this way closed control loop control system is called **Automatic control system**.
- Some example:
 - Automatic Electric Iron,
 - Water Level Controller
 - An air Conditioner



Block Diagram of Open Loop System



Air conditioner



Geyser



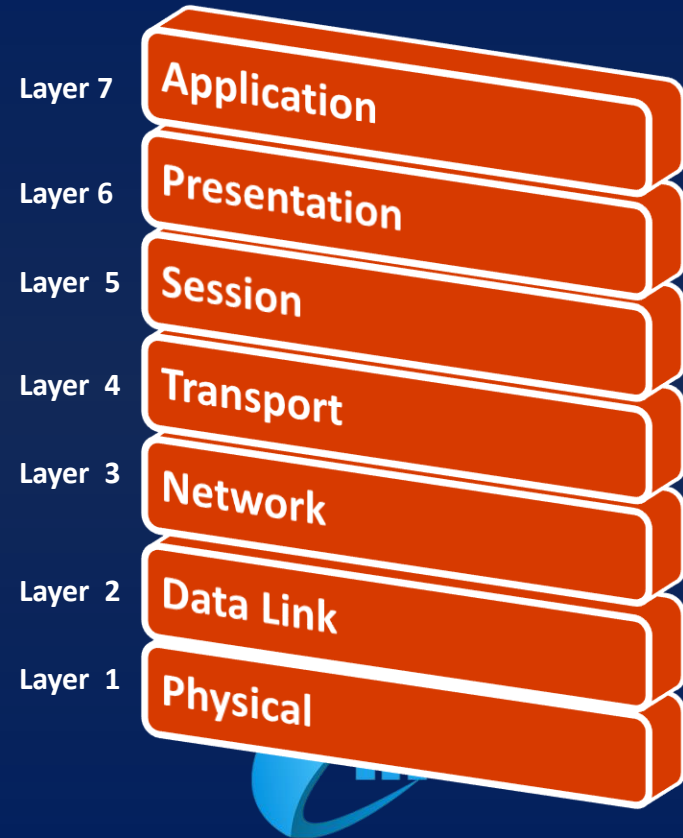
Connectivity Model

- It would be useful to have a set of protocols and rules for data transmission for the variety of network types to ensure that the information can be shared.
- For this reason, OSI and TCP/IP models were created.
- These models describe network system as being made up of a number of layers, each with its own set of protocols and functionality.
 - OSI Model
 - TCP/IP Model



OSI (Open System Interconnection) Model

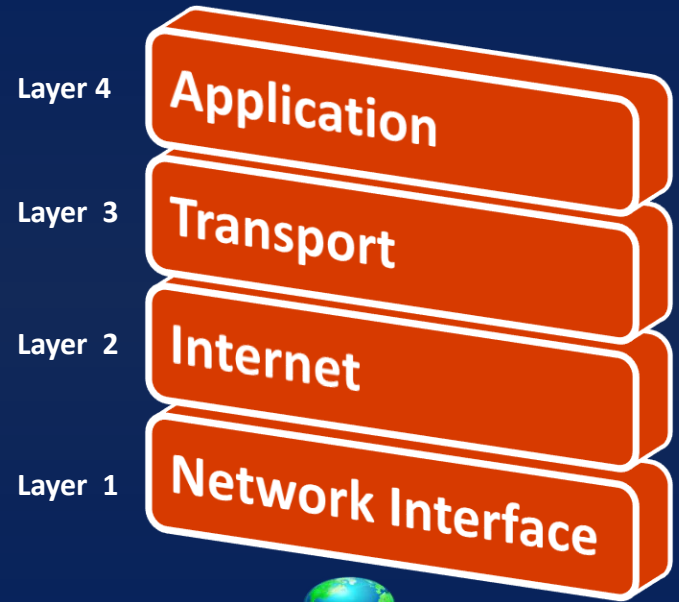
- The Open System Interconnection (OSI) model is the foundation of internet communications, including for devices connected to the IoT.
- The OSI model was developed to standardize the procedure for exchange of information between processing system.
- It is developed by **ISO (International Standard Organization)**



OSI Layer	Function	Common Protocols	Examples / Applications
Application	Interface for end-user processes, network services to applications	HTTP, FTP, SMTP, POP3, IMAP, DNS, Telnet	Web browsers, Email clients, File transfer tools
Presentation	Data translation, encryption, compression	SSL/TLS, JPEG, MPEG, GIF	SSL in HTTPS, Data format converters
Session	Establishes, manages, and terminates sessions	NetBIOS, PPTP, RPC	Remote procedure calls, Session management APIs
Transport	Reliable or unreliable data delivery, error recovery	TCP, UDP, SCTP	TCP (web traffic), UDP (streaming, VoIP)
Network	Logical addressing, routing, path determination	IP, ICMP, IGMP, IPSec, RIP, OSPF	Routers, IP addressing, VPNs
Data Link	Physical addressing, error detection/correction	Ethernet (IEEE 802.3), PPP, HDLC, ARP, Frame Relay	Switches, MAC addresses, Network Interface Cards
Physical	Transmission and reception of raw bit streams	Ethernet (physical), USB, RS-232, DSL	Cables, Hubs, Repeaters, Network adapters

TCP/IP Model

- TCP/IP (Transmission Control Protocol/Internet protocol) is based on a client-server communication model by which a user (client) receives a service such as loaded a web page from server in the same network.
- TCP/IP is an industry-standard protocol suite for Wide Area Networks (WANs) developed in the 1970s and 1980s by the U.S. Department of Defence (DoD).
- Initially it is known as ARPAnet (Advanced Research Project Agency Network).



TCP/IP Layer	Function	Common Protocols	Examples / Applications
Application	High-level APIs, network services, data formatting	HTTP, HTTPS, FTP, SMTP, POP3, DNS, DHCP, SNMP	Web browsers, Email clients, File transfer apps
Transport	End-to-end communication, error handling, flow control	TCP, UDP, SCTP	TCP (reliable delivery), UDP (real-time apps)
Internet	Logical addressing, routing, packet forwarding	IP (IPv4/IPv6), ICMP, IGMP, ARP	Routers, IP addressing, Path selection
Network Access / Link	Physical transmission, MAC addressing, hardware interfaces	Ethernet, Wi-Fi, PPP, Frame Relay, ARP	Switches, Network cards, Physical media

Type of Modes

- There are three type of data transmission that correspond to the three types of circuits available.

These modes are:

- Simplex
- Half-duplex
- Full-duplex



Transmission Media

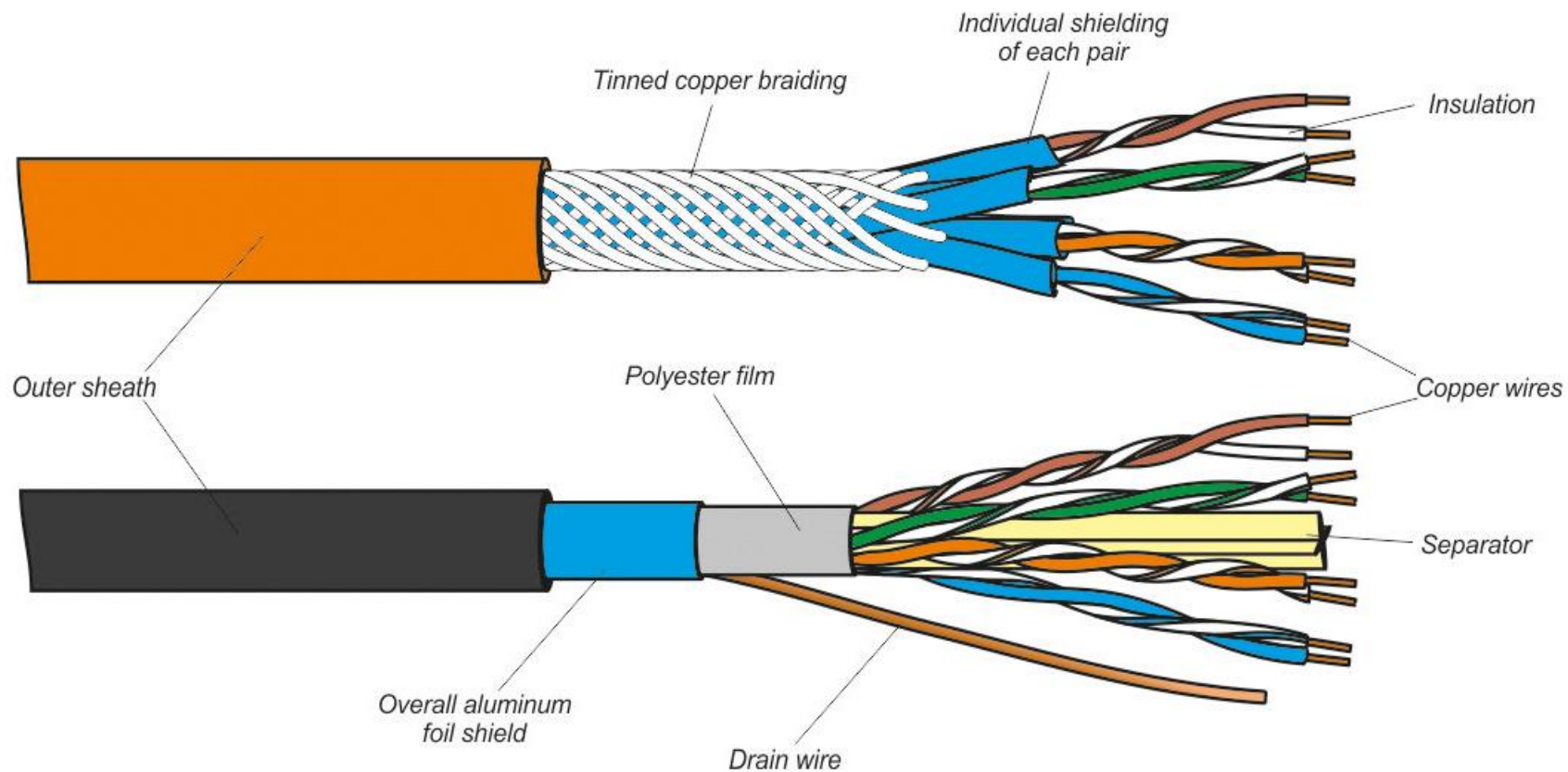
- There are two categories of Transmission Media:
 - Guided Media
 - Unguided Media
- Guided Media: Guided media follow the fixed path for transmission.
 - Example: Twisted-pair Cable, Coaxial Cabling, Fiber-optic cabling
- Unguided Media : Media in which the signal are not guided through solid medium are known as Unguided Media.
 - Example: Radio waves, Micro Waves, Infrared Waves



Twisted-pair Cable

- A twisted pair of wires consists of two insulated copper wires, typically about 1 mm thickness.
- The wires are twisted together in a helical shape.
- The purpose of twisting the wires is to reduce electrical interference.
- Twisted pair wire are commonly used in local telephone communication and for digital data transmission over short distance up to 1 Km.
- Twisted pair cabling, are of two types:
 - Unshielded twisted pair
 - Shielded twisted pair

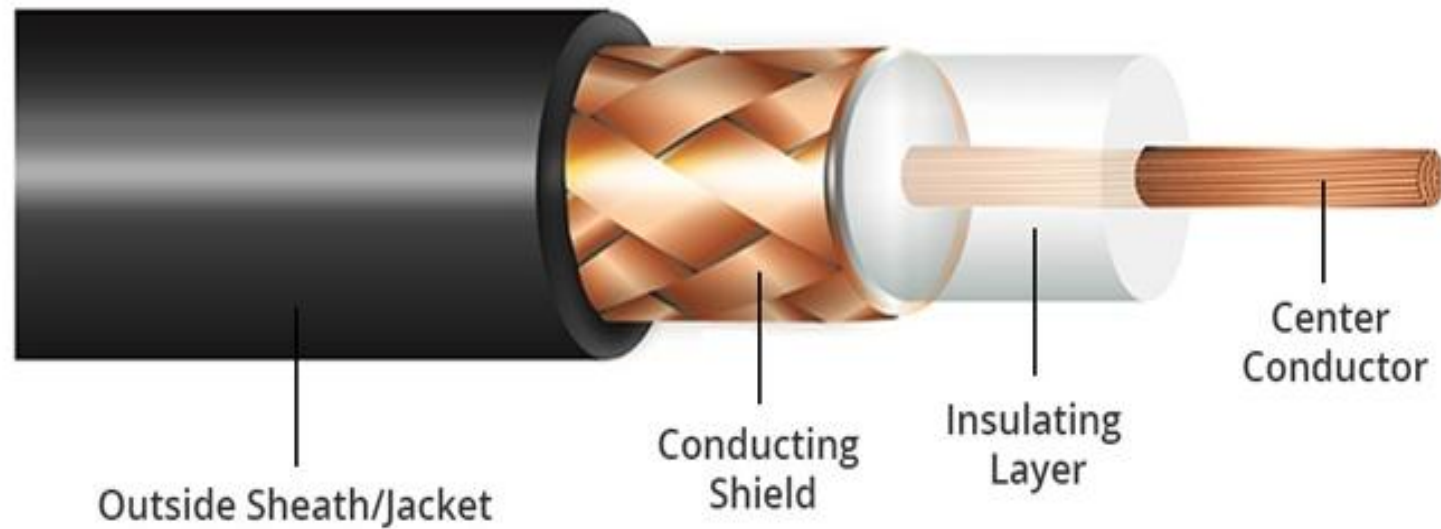




Coaxial Cable

- Coaxial cables carry the signal of higher frequency range compared to twisted pair cable.
- Coaxial cables are also called coax (short form)
- There are two types of coaxial cables widely used : 50 ohm and 75 ohm cable.
- 50 ohm cable is used for digital transmission and 75 ohm cable used for analog transmission.

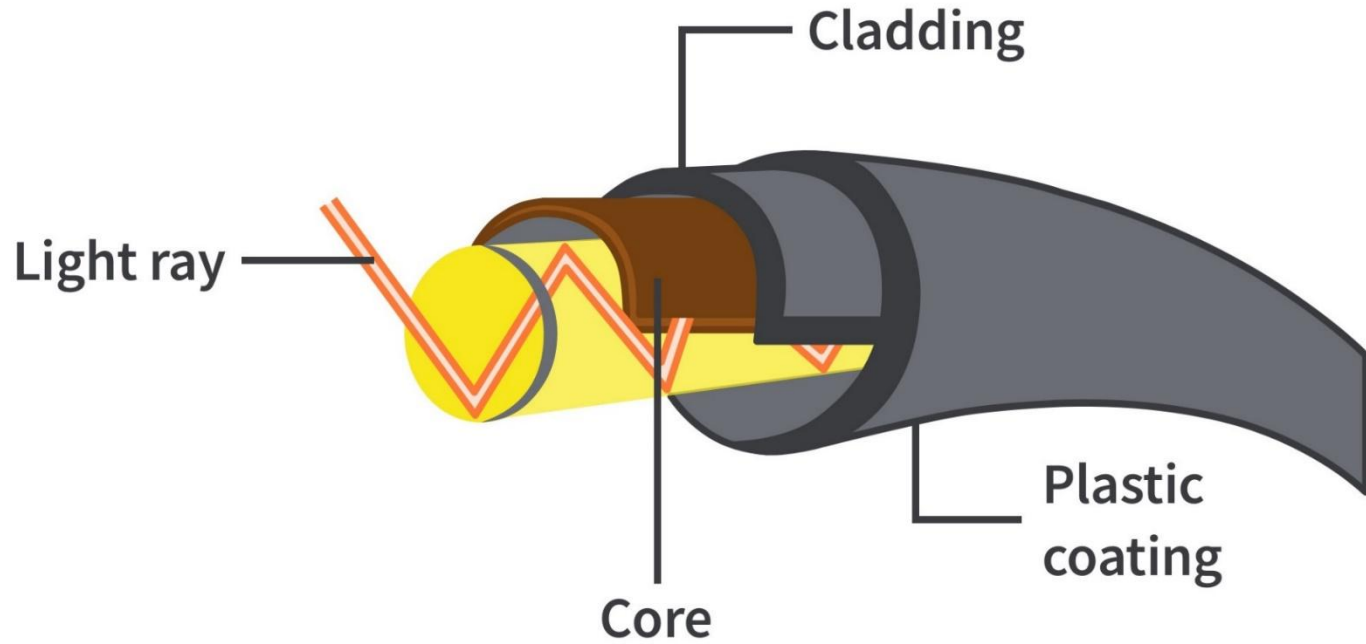




Fiber Optic Cable

- Fiber optic cable is made of glass strands that transmit signals as light waves or pulses.
- Each fiber is surrounded by a plastic sheath.
- Fiber optic cable is generally used for backbone for high speed interconnection between networking devices.
- Fiber optic cable can easily handle data at speeds above 1Gbps, it has been demonstrated to handles above data rate exceeding 200Gbps.
- It can handle much higher bandwidth than copper.
- Fiber is lighter than copper.





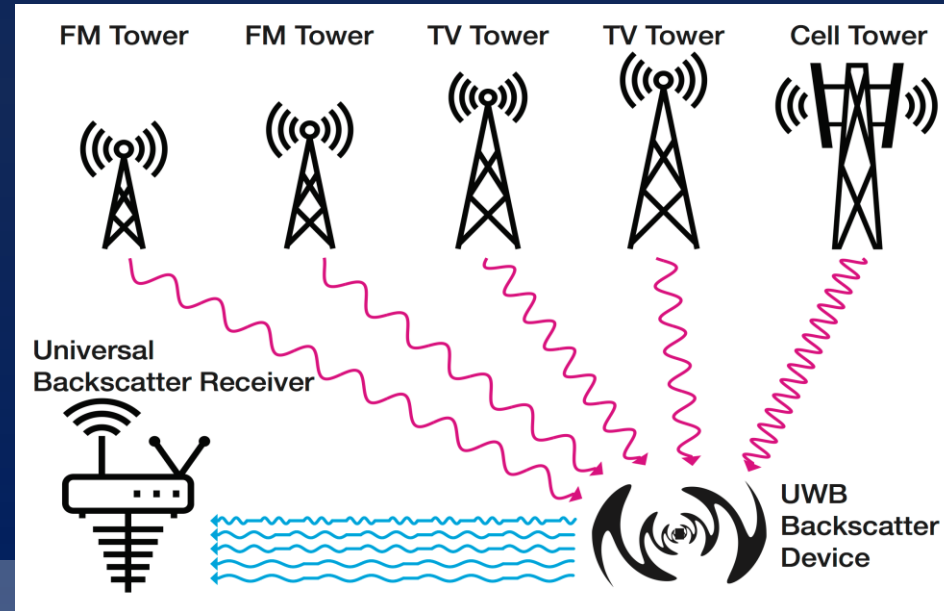
Unguided media

- Media in which the signals are not guided through a solid medium are known as unguided media.
- Air is the media through which electromagnetic energy can flow easily.
- Therefore, there are several methods which are in use to send electromagnetic energy through air.
- These methods are:
 - Radio waves
 - Micro waves
 - Infrared Waves



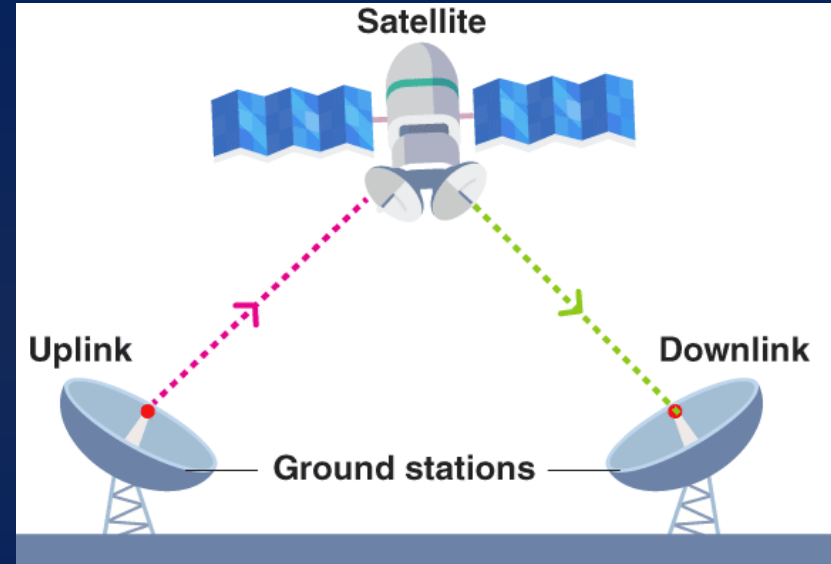
Radio waves

- Electromagnetic waves ranging in frequencies between 10 Kilohertz and 1 gigahertz.
- Radio waves can be broadcast Omni directional or unidirectional.
- Various kind of antennae can be used to broadcast radio signals.
- Radio waves are easy to generate.
- They can travel long distance.



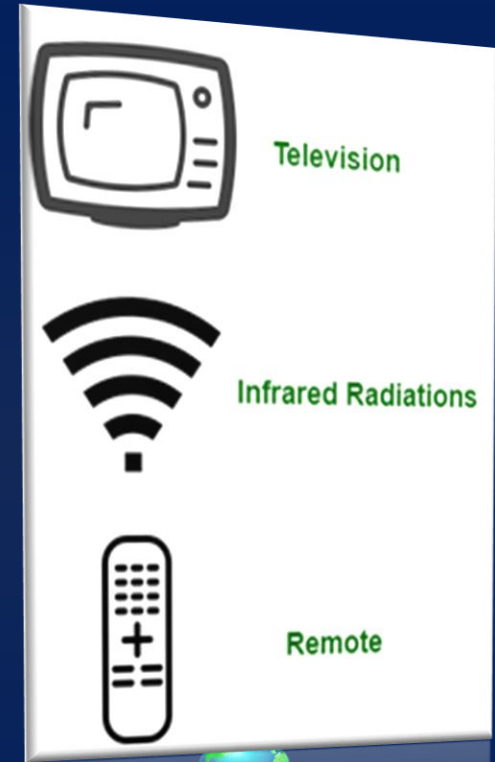
Microwave

- Above 100 MHz, the waves travel in straight lines and can therefore be narrowly focused.
- Concentrating all the energy into a small beam using a parabolic antenna (like the satellite TV dish) gives much higher signal to noise ratio.
- They are used in Cellular phones, satellite networks and wireless LANs.



Infrared

- Unguided infrared and millimetre waves are widely used for short-range communication.
- The remote controls used on television, VCRs used infrared communication.
- They are relatively directional, cheap and easy to build but do not pass through solid objects.
- Infrared communication is used for indoor wireless LANs.



Thank You

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